

LAB EXPERIMENT RECORD

Set 4: Product Inventory Management

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

Aim

To visualize product inventory data using a bar chart, stacked bar chart, and scatter plot, and to develop an interactive Tableau dashboard.

PROGRAM 1: Bar Chart of Product Inventory

Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

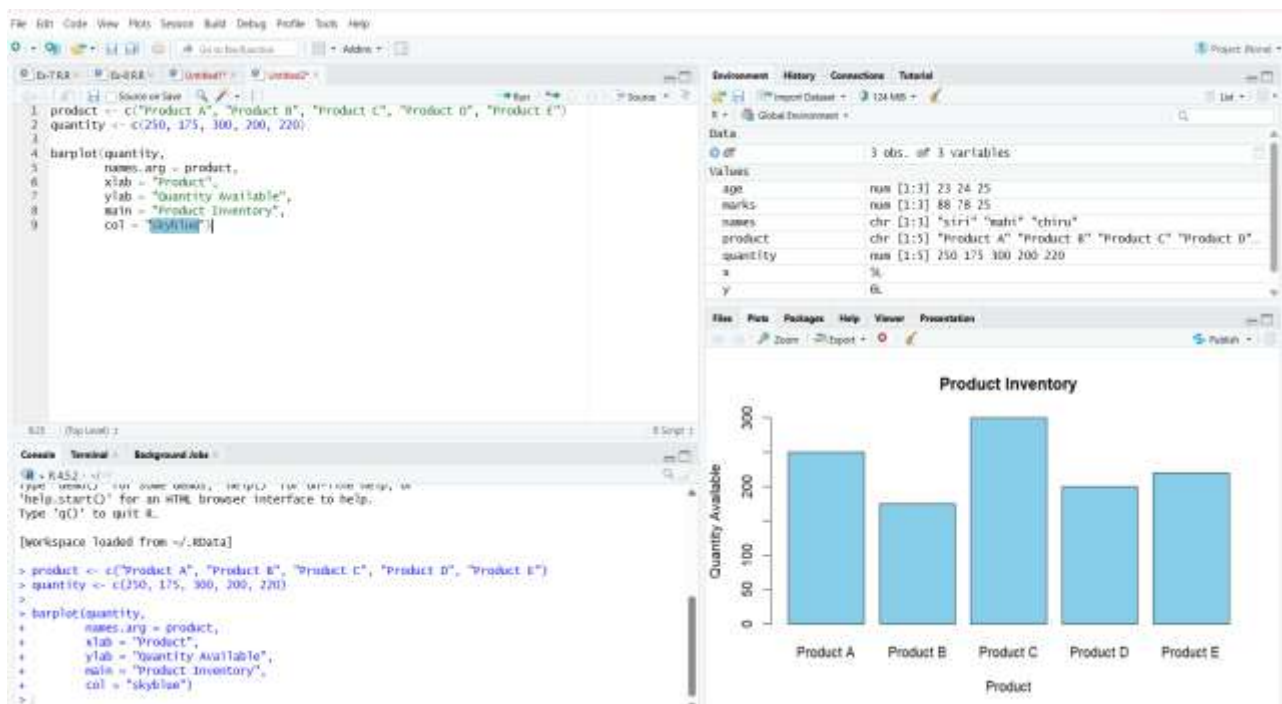
```
product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
```

```
quantity <- c(250, 175, 300, 200, 220)
```

```

barplot(quantity,
        names.arg = product,
        xlab = "Product",
        ylab = "Quantity Available",
        main = "Product Inventory",
        col = "skyblue")

```



Observation: The bar chart represents the quantity available for the five products. Product C has the highest quantity of 300, while Product B has the lowest quantity of 175.

PROGRAM 2: Stacked Bar Chart by Product Category

Generate a stacked bar chart to show the quantity of each product within different product categories.

```

product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
category <- c("Category 1", "Category 1", "Category 2", "Category 2", "Category 3")
quantity <- c(250, 175, 300, 200, 220)
data <- table(product, category)
barplot(t(data * quantity),

```

```

beside = FALSE,

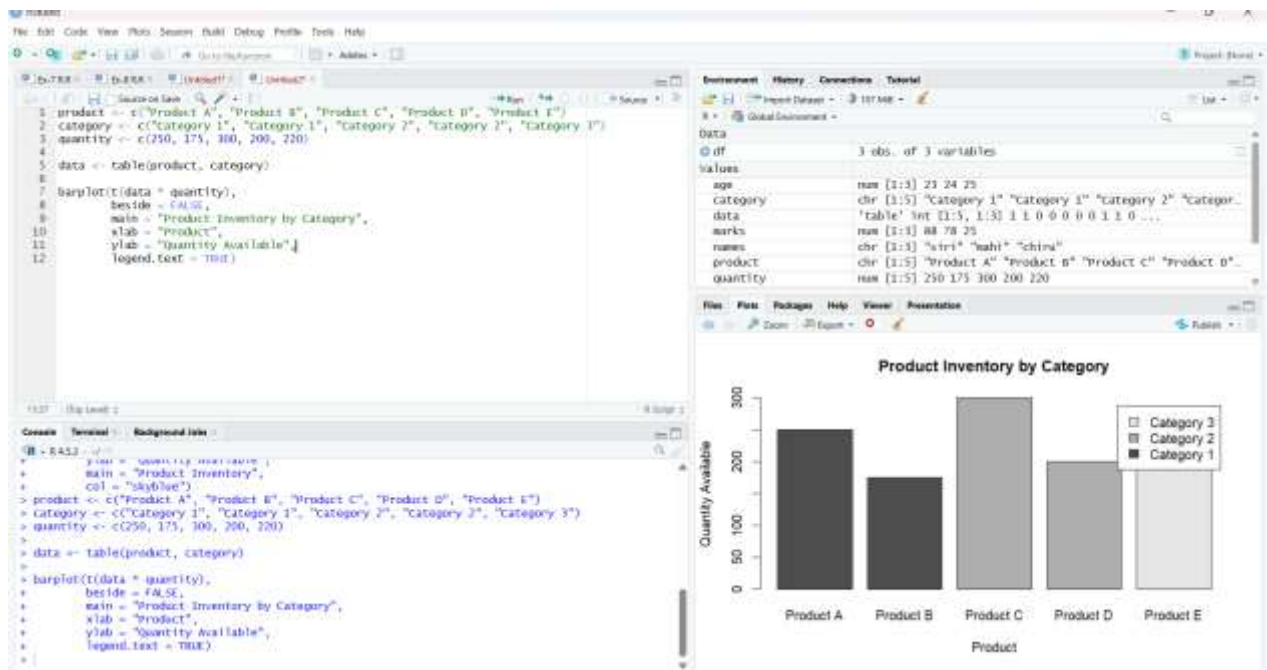
main = "Product Inventory by Category",

xlab = "Product",

ylab = "Quantity Available",

legend.text = TRUE)

```



Observation: The stacked bar chart shows the quantity of products according to their respective categories. It helps compare product quantities while also showing the category associated with each product.

PROGRAM 3: Scatter Plot of Product Price and Quantity Available

Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

```

price <- c(100, 150, 80, 120, 110)

quantity <- c(250, 175, 300, 200, 220)

plot(price,

      quantity,

      main = "Product Price vs Quantity Available",

      xlab = "Product Price",

      ylab = "Quantity Available",

```

```

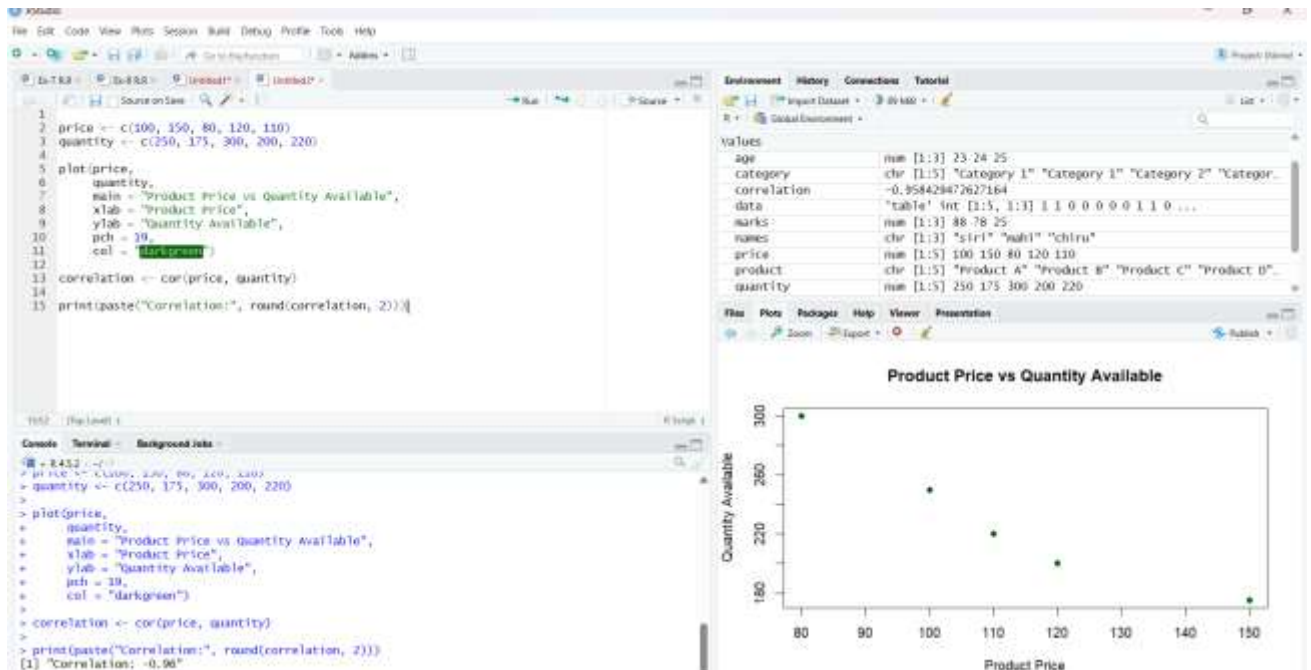
pch = 19,

col = "darkgreen")

correlation <- cor(price, quantity)

print(paste("Correlation:", round(correlation, 2)))

```



Findings: The scatter plot shows the relationship between product price and quantity available. For the sample values used, products with higher prices generally have lower quantities available, indicating a negative relationship between price and quantity.

PROGRAM 4: Product Inventory Tableau Dashboard

Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

```

library(shiny)

product <- c("Product A","Product B","Product C","Product D","Product E")

quantity <- c(250,175,300,200,220)

ui <- fluidPage(

  titlePanel("Product Inventory Dashboard"),

  selectInput("chart","Select Chart:",

              choices=c("Bar Chart","Stacked Bar Chart")),

  plotOutput("plot")

```

```

)

server <- function(input, output) {

  output$plot <- renderPlot({

    if(input$chart == "Bar Chart") {

      barplot(quantity,

        names.arg=product,

        xlab="Product",

        ylab="Quantity Available",

        main="Product Inventory")

    } else {

      barplot(quantity,

        names.arg=product,

        xlab="Product",

        ylab="Quantity Available",

        main="Product Inventory by Category")

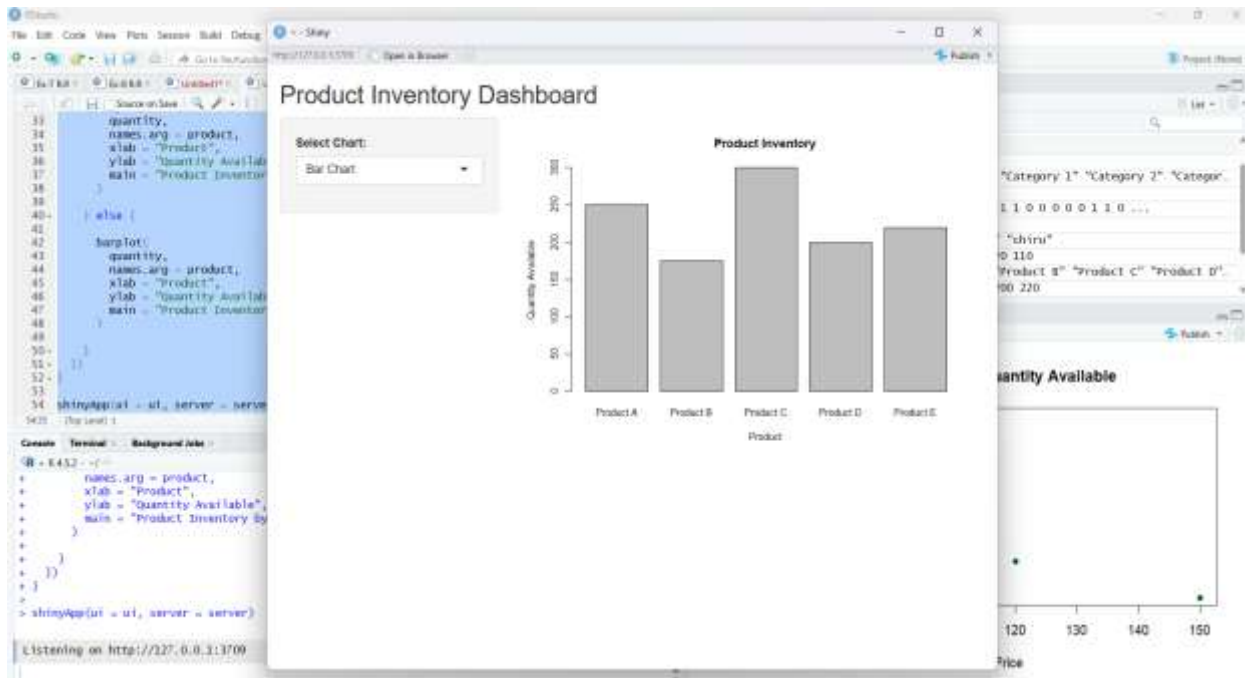
    }

  })

}

shinyApp(ui, server)

```



Observation

The dashboard provides a chart-selection control. Selecting Bar Chart displays the quantity of each product, while selecting Stacked Bar Chart displays the product quantities according to their categories.